

Mud Mortar Improvement Through the Addition of Fibres

Joint Research Project

CRS – Oxford Brookes University – Purdue University

The Problem

- HHs are building in a semi-traditional way, not taking advantage of hazard resistant elements that have developed over millennia
- Appropriate or traditional materials are becoming more difficult to source and having a negative impact on the environment
- Mud mortar has a low tensile strength and can crumble easily
- Lime and cement mortar can improve the strength of mortar
- Access to and cost of cement and lime is prohibitive
- Taking advantage of some of the inherent properties of mud, how can we improve its strength?
- The use of by products or waste to improve strength could provide other advantages

First Round of testing with OBU

- Test samples were provided in grey cement mortar and traditional mud mortar with the following types of reinforcement:
 - None (control)
 - Rope
 - Straw
 - Shredded tarpaulin
- Two types of test were undertaken:
 - Test to initial failure (drop in load)
 - Test to total failure (sample breaks in two)



Round 1 results

- Although the reinforcement did little to improve the shear resistance of the mortar, the improved post-cracking behaviour is likely to have some benefit in terms of the earthquake resilience of the masonry, by improving the structural integrity of the building and reducing the risk of total collapse during a seismic event.



Second Round of Testing with OBU

- Fibres commonly used in earth mortar include straw, hay, hemp, sisal, and elephant grass. This type of fibre was tested, along with man-made fibres which might commonly be available for low or no cost in a post disaster context: rope and tarpaulin.
- The test programme is divided into three phases as follows:
 - Phase 1 - shear tests to investigate impact of fibre density and length
 - Phase 2 - additional shear tests with alternative reinforcement options
 - Phase 3 – Static and dynamic testing of 1m x 1m wall samples

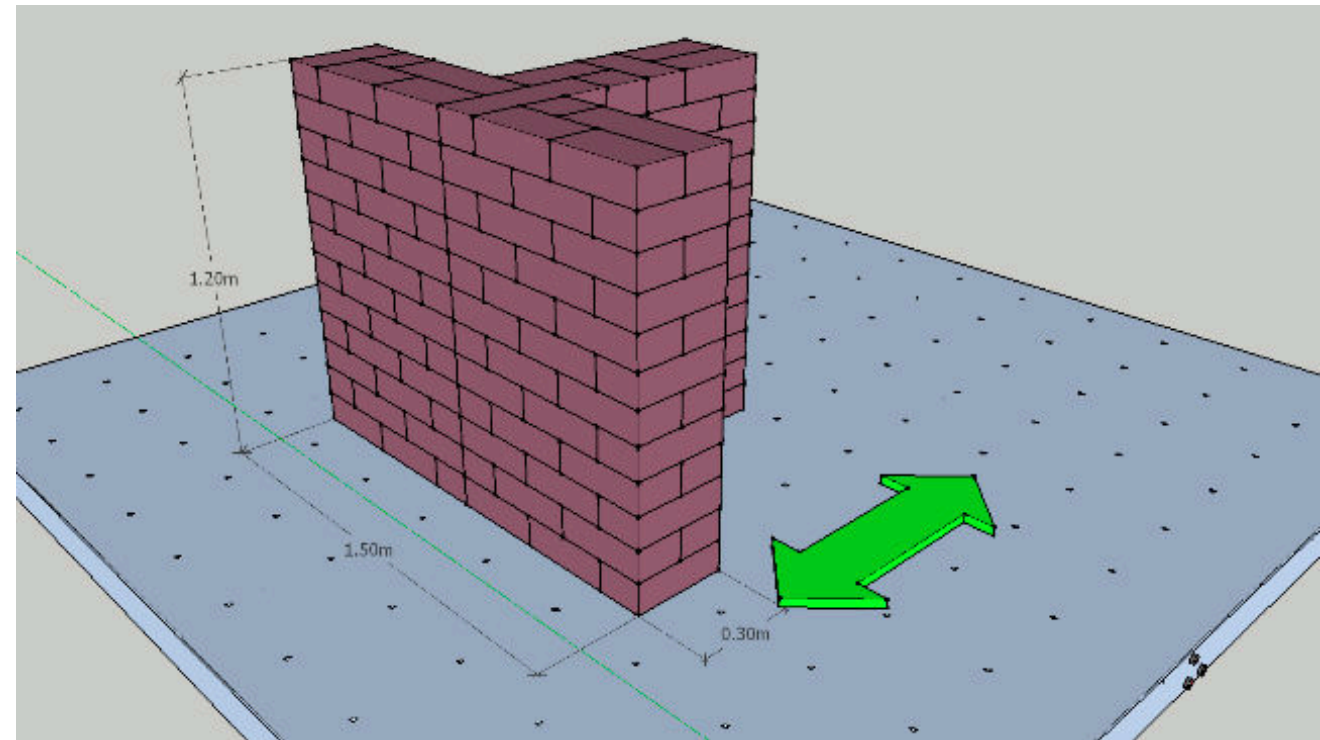




Round 3 Testing with Purdue University

Dynamic Tests of Stone Masonry Walls with Earth Mortar

- T-shaped stone masonry wall systems representative of typical interior-to-exterior wall joints will be built and tested on the earthquake simulator at Purdue University. Two specimens shall incorporate improved earth mortar currently in development at Oxford Brookes University (OBU) and the second without OBU's proposed improvements.
- Type N: No mortar (dry stone)
- Type W: Earth mortar with fibers and NSET detailing recommendations (e.g., wood banding)
- Type I: Other improvements





Thank you!

